

SMOLENSK, Russian Federation

WMO: 267810

Lat: 54.7491N

Lon: 32.0647E

Elev: 238

StdP: 98.50

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.0	-18.8	-24.6	0.4	-21.5	-21.6	0.6	-18.1	8.4	-1.1	7.3	-2.8	1.5	90	0.542

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	28.1	20.1	26.4	19.1	24.7	18.3	20.9	26.3	20.0	25.1	19.1	23.6	2.4	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.1	14.2	23.7	18.1	13.4	22.5	17.3	12.7	21.5	61.6	26.4	58.2	25.1	55.2	23.5	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.7	5.9	5.2	-25.9	30.7	2.9	2.2	-28.0	32.2	-29.8	33.5	-31.4	34.7	-33.6	36.3		
(5)			DB	WB	-26.1	22.3	2.9	1.0	-28.2	23.1	-29.9	23.7	-31.5	24.2	-33.6	25.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.9	-6.4	-5.9	-1.0	6.8	12.6	16.0	18.6	16.7	11.4	5.7	0.1	-4.4
	DBStd	9.97	6.03	6.28	4.52	4.82	4.61	3.49	3.12	3.40	3.61	4.02	4.77	5.93
	HDD10.0	2356	509	446	342	120	26	1	0	0	26	142	298	446
	HDD18.3	4628	768	679	600	346	186	87	36	73	209	392	548	704
	CDD10.0	862	0	0	0	24	106	181	265	209	68	9	0	0
	CDD18.3	90	0	0	0	0	7	16	43	23	1	0	0	0
	CDH23.3	694	0	0	0	3	61	109	318	191	12	0	0	0
	CDH26.7	139	0	0	0	0	6	11	71	51	1	0	0	0
	WSAvg	2.4	2.8	2.7	2.7	2.5	2.3	2.1	1.9	1.9	1.9	2.4	2.7	2.8
(15) Precipitation	PrecAvg	674	43	34	40	42	50	71	95	67	64	58	58	56
	PrecMax	877	87	87	86	79	92	123	191	184	226	140	109	90
	PrecMin	508	8	6	5	8	14	12	4	18	12	11	29	21
	PrecStd	92	23	19	20	21	21	31	47	36	46	30	23	17
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.7	4.7	13.3	23.0	27.4	28.1	31.1	31.6	25.0	18.8	11.1	7.9
		MCWB	4.0	3.3	7.6	13.0	18.7	19.7	21.3	20.5	18.0	13.4	10.0	7.1
	2%	DB	2.8	3.0	9.4	19.9	25.2	26.3	28.6	27.5	21.6	14.5	8.8	5.4
		MCWB	2.0	1.8	4.6	11.4	17.0	19.2	20.4	19.7	15.7	12.1	7.9	4.8
	5%	DB	1.6	2.0	6.9	17.1	22.6	24.5	26.7	25.2	19.3	12.9	7.5	3.6
		MCWB	1.0	1.1	3.5	10.4	15.5	18.2	19.8	18.5	14.6	11.0	6.6	3.0
	10%	DB	0.6	1.1	4.8	14.8	20.4	22.6	24.9	23.1	17.2	11.3	6.1	2.0
		MCWB	0.2	0.4	2.4	9.1	14.3	17.2	18.8	17.4	13.4	9.8	5.4	1.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.3	3.7	7.8	13.9	19.9	21.1	22.6	21.7	18.6	14.3	10.4	7.3
		MCDB	4.8	4.4	12.1	20.9	26.3	26.0	28.7	27.9	22.9	16.2	11.1	7.7
	2%	WB	2.2	2.1	5.4	12.1	18.0	19.9	21.3	20.5	16.5	12.7	8.2	5.0
		MCDB	2.6	2.6	8.4	17.9	23.5	24.7	26.8	26.2	20.3	14.3	8.6	5.3
	5%	WB	1.0	1.2	3.9	11.0	16.5	19.0	20.3	19.4	15.0	11.2	6.7	3.2
		MCDB	1.4	1.7	6.0	16.2	21.1	23.4	25.3	24.1	18.1	12.6	7.3	3.6
	10%	WB	0.2	0.6	2.7	9.8	15.2	17.8	19.4	18.2	13.9	10.1	5.5	1.6
		MCDB	0.6	1.1	4.5	14.1	19.2	21.7	24.0	22.0	16.6	11.1	6.0	1.9
(35) Mean Daily Temperature Range	5% DB	MDBR	4.6	5.5	6.6	8.7	9.6	8.9	9.3	9.4	8.6	6.2	4.1	4.0
		MCDBR	3.4	4.2	9.2	12.0	12.7	11.3	11.8	12.5	11.9	8.6	5.4	3.7
		MCWBR	3.5	3.8	5.6	5.8	6.0	5.5	5.2	5.6	6.4	5.7	4.7	3.5
	5% WB	MCDBR	3.4	3.9	7.8	11.0	11.0	10.2	10.8	11.1	10.3	7.3	4.8	3.7
		MCWBR	3.6	3.7	5.3	5.5	5.8	5.5	5.4	5.5	6.1	5.5	4.4	3.6
(40) Clear-Sky Solar Irradiance	taub		0.286	0.320	0.345	0.407	0.391	0.372	0.400	0.414	0.369	0.346	0.325	0.288
	taud		2.145	2.091	2.143	2.170	2.309	2.379	2.317	2.272	2.371	2.392	2.256	2.161
	Ebn at Noon		631	735	814	812	853	875	838	799	792	720	584	539
	Edh at Noon		63	96	115	130	119	113	118	116	92	72	58	51
(44) All-Sky Solar Radiation	RadAvg		0.66	1.55	2.99	4.00	5.26	5.64	5.44	4.38	2.91	1.38	0.51	0.39
	RadStd		0.15	0.25	0.38	0.44	0.41	0.52	0.50	0.49	0.27	0.28	0.10	0.11

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.65	N/A	N/A	N/A	N/A	N/A	N/A	-210	N/A	N/A
(47)	Regional (8 neighbors)	+0.62	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page